

Image Fundamentals

Reference:

[Gonzalez and Woods] Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Prentice-Hall, Inc.

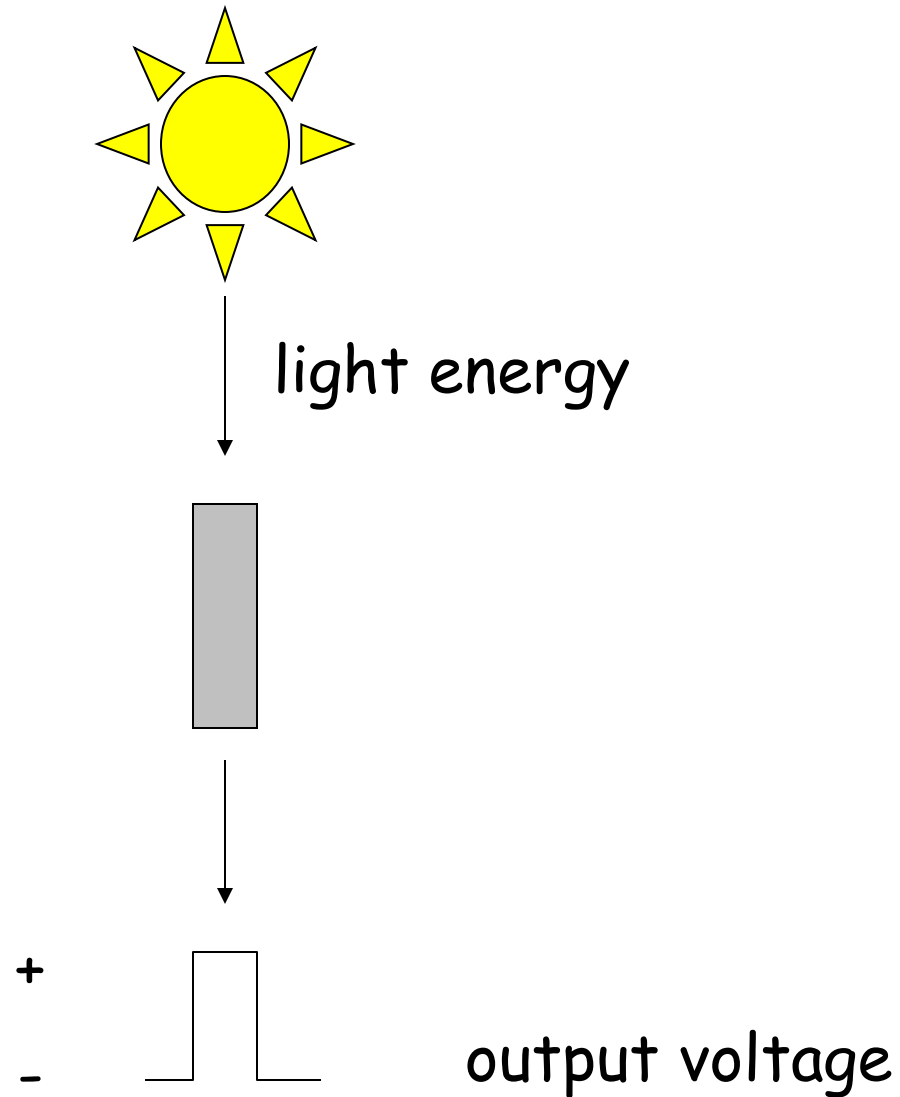
<http://www.imageprocessingplace.com/>

Digital Imaging

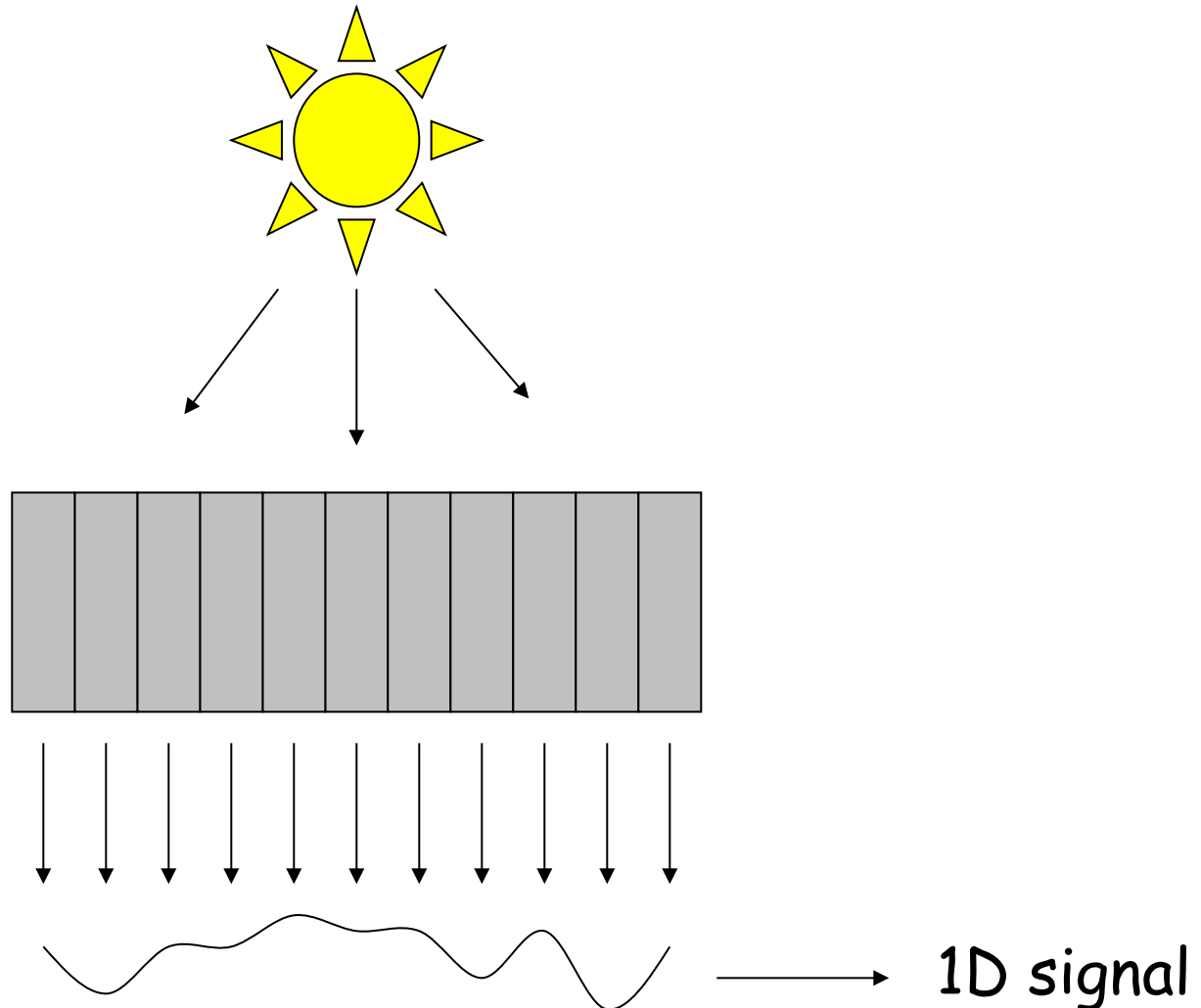
- Charged-Coupled Device (CCD)
 - consists of photosites (regions for collecting light)
 - Each photosite (region) is a silicon imaging element (sensor) that gives a voltage output proportional to the intensity of the incident light
 - linear array (scanner)
 - area array (Camera CCD)



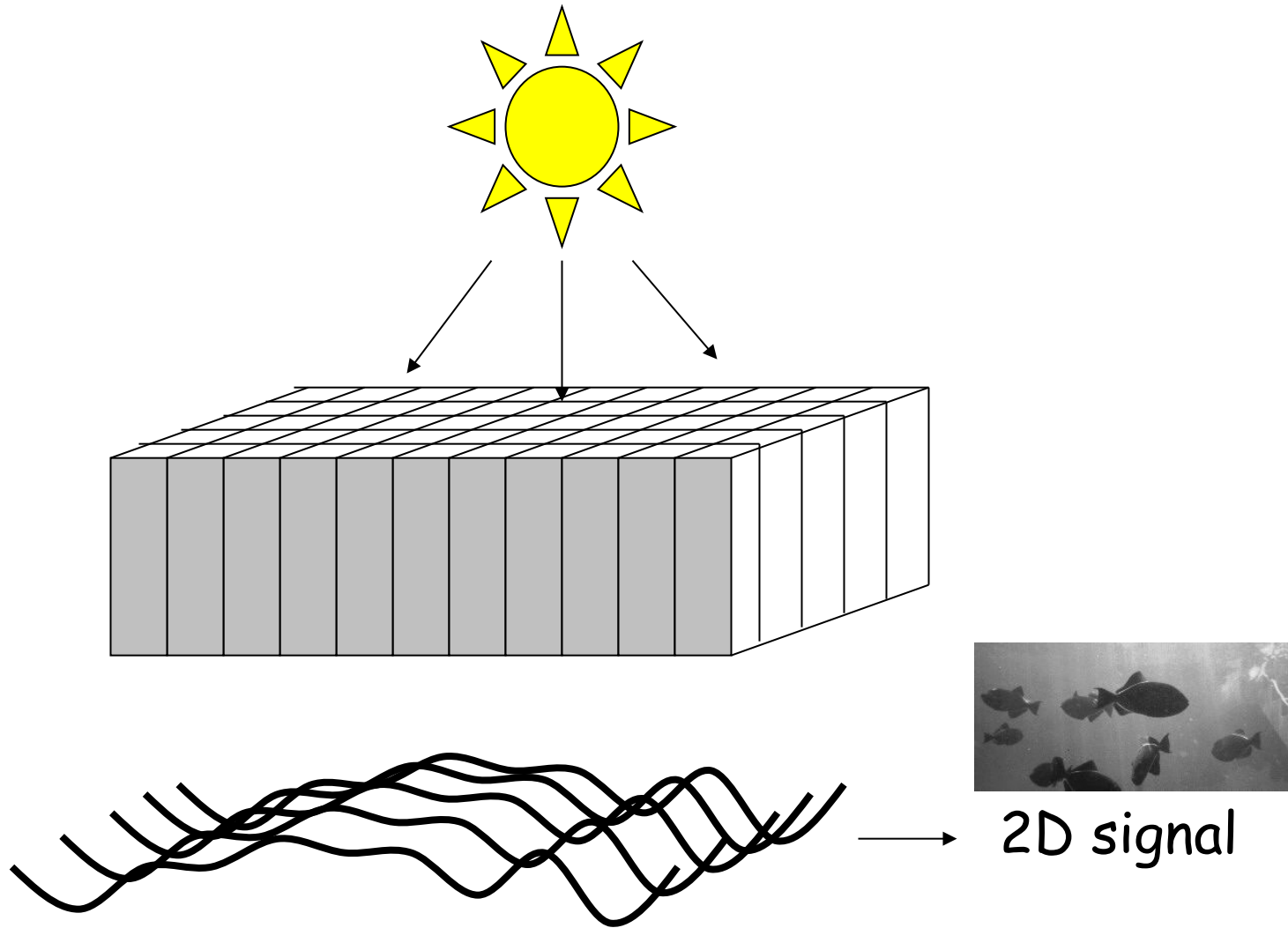
Photosite



Line scan sensor = a line of photosites



Area scan sensor = 2D array of photosites



Photosite (Single Sensing Element)

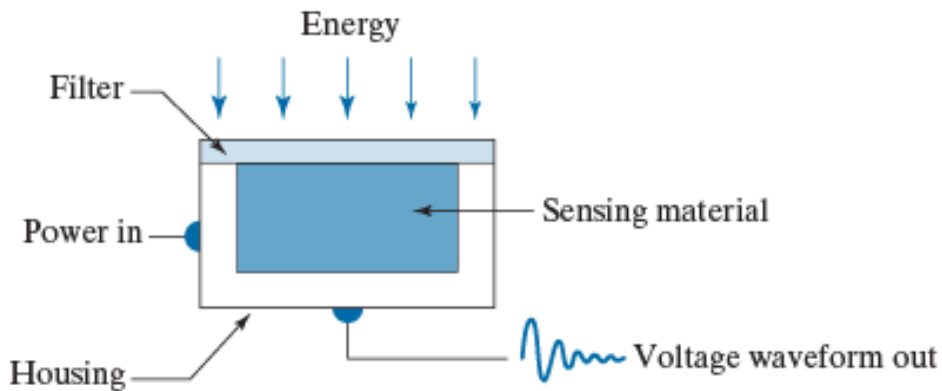


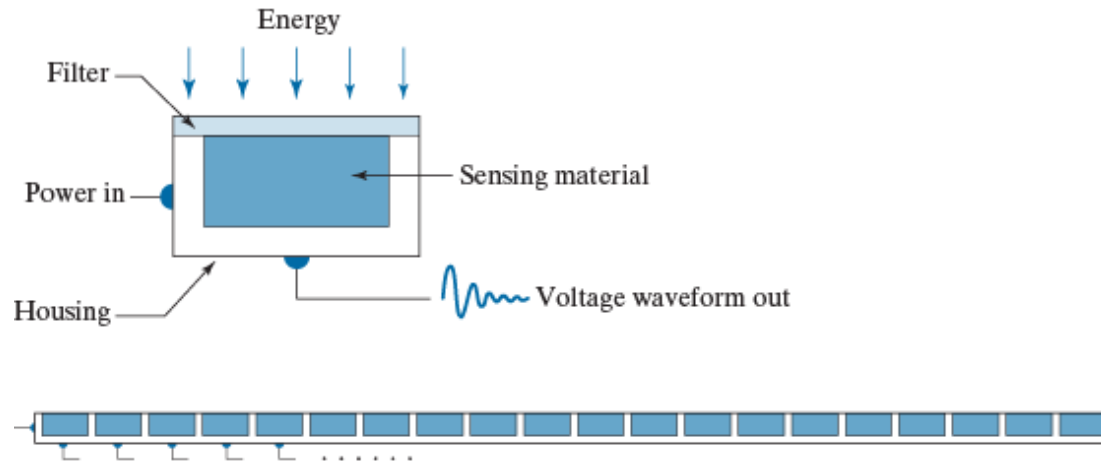
FIGURE 2.12
(a) Single sensing element.



Photodiode

[Gonzalez and Woods]

Line scan sensor = a line of photosites
2D sensor = multiple lines of photosites



a
b
c

FIGURE 2.12

- (a) Single sensing element.
- (b) Line sensor.
- (c) Array sensor.

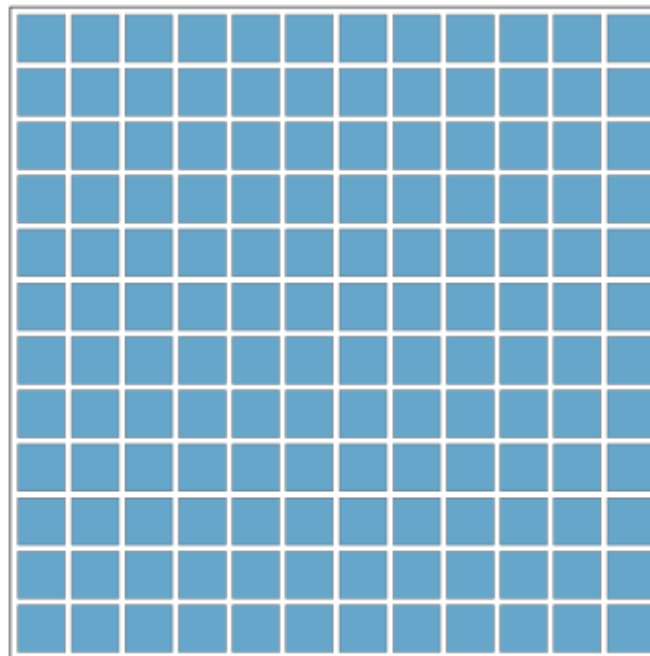
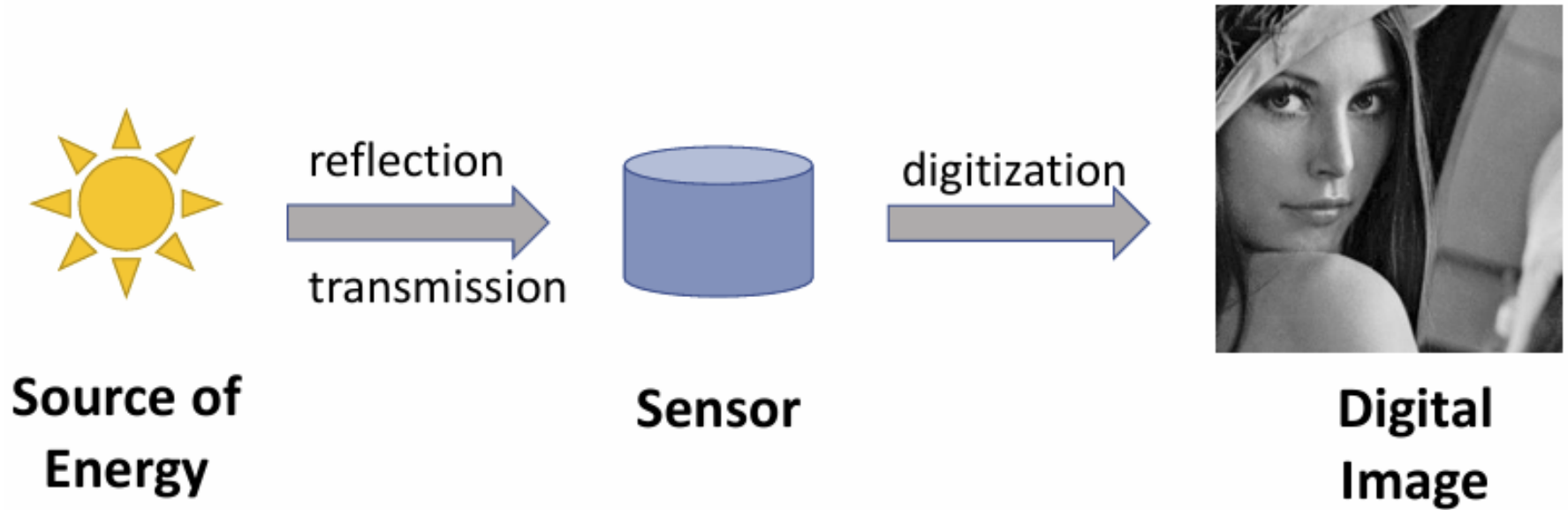
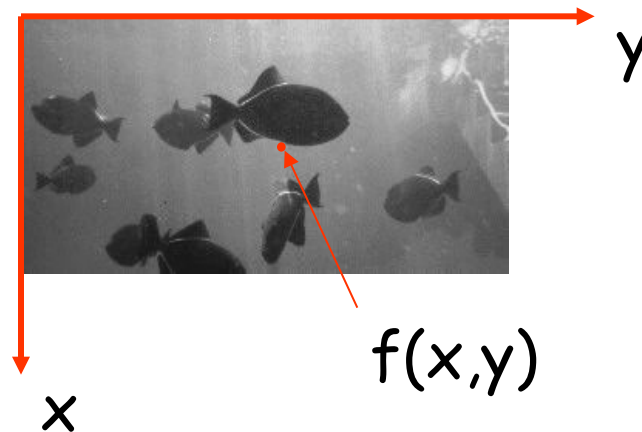


Image Acquisition



Continuous image function

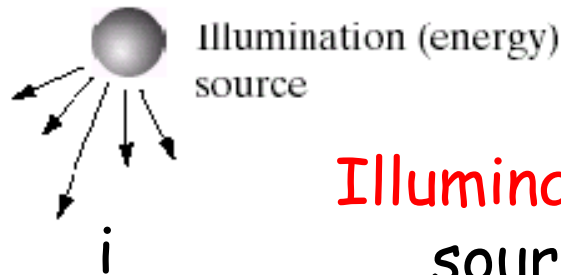


The continuous image function (intensity/color) f can be characterized by two components:

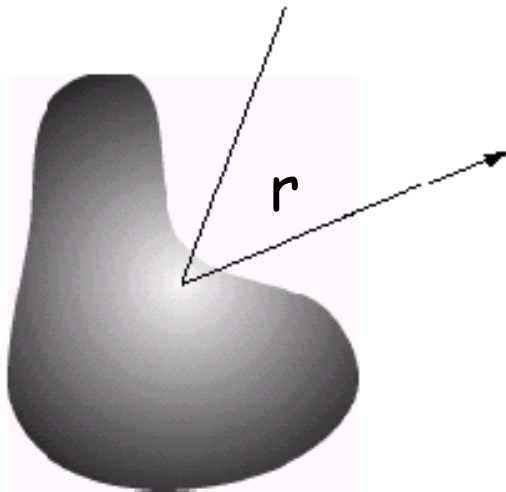
- (1) illumination (light source) and
- (2) reflectance (materials).

Image intensity depends on these two factors.

Continuous image function



Illumination: the amount of source illumination (energy) **incident** on the scene being viewed.



Scene element

Reflectance: the amount of source illumination (energy) **reflected** by the objects (or elements) in the scene.

$$f(x,y) = i(x,y) r(x,y)$$

Continuous image function

- Basic nature of $f(x,y)$ has two components:

$$f(x,y) = i(x,y) r(x,y)$$

- $i(x,y)$ is illumination component

$$0 \leq i(x,y) < +\infty$$

- $r(x,y)$ is reflectance component

$$0 \leq r(x,y) \leq 1$$

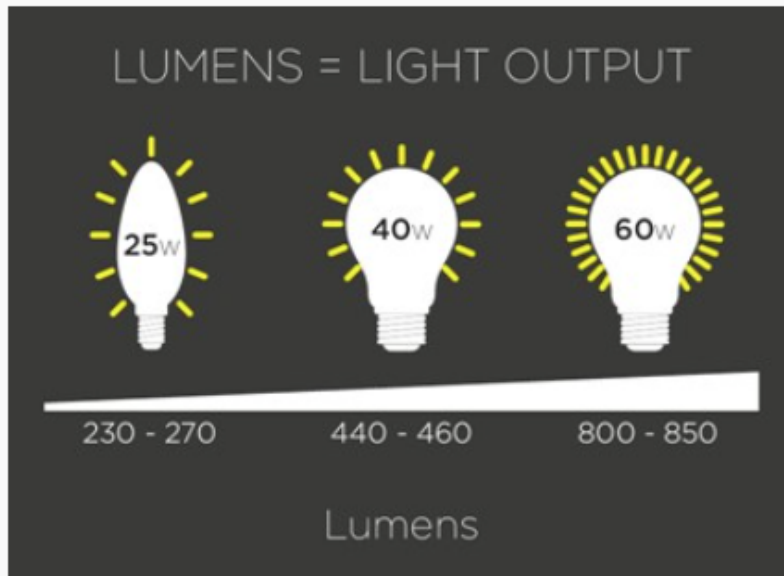
0 = total absorption, 1 = total reflectance.

- $f(x,y)$ is intensity

- $0 \leq f(x,y) < +\infty$

Lumens

What are Lumens?



Think of lumens as a "new" way of knowing how bright a lamp is.
Lumens = Light Output.

In simple terms, Lumens (denoted by lm) are a measure of the total amount of visible light (to the human eye) from a lamp or light source. The higher the lumen rating the "brighter" the lamp will appear.

Illumination: Typical Values



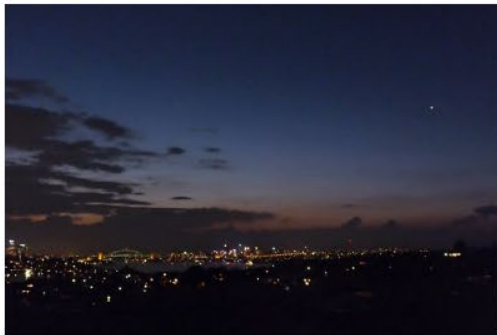
Sunny day

$90,000 \text{ lm/m}^2$



Cloudy day

$10,000 \text{ lm/m}^2$



Night

0.1 lm/m^2



Office

$1,000 \text{ lm/m}^2$

lm: Lumen.
Unit for illumination

Photo by Diana Evans from FreemImages

<https://lovepik.com/download/detail/501034484?byso=&type=0>

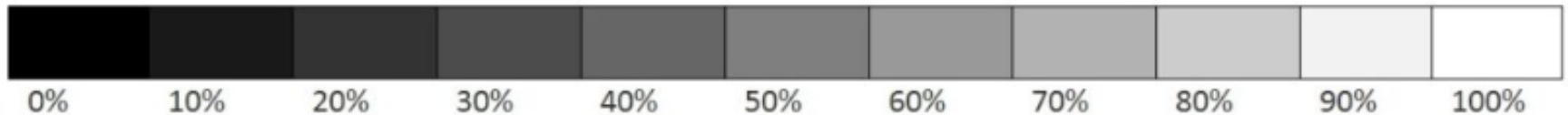
<https://www.westend61.de/en/imageView/CUF49861>

<https://nightinfoocus.wordpress.com/tag/evening-star/>

Light Reflective Value (LRV)

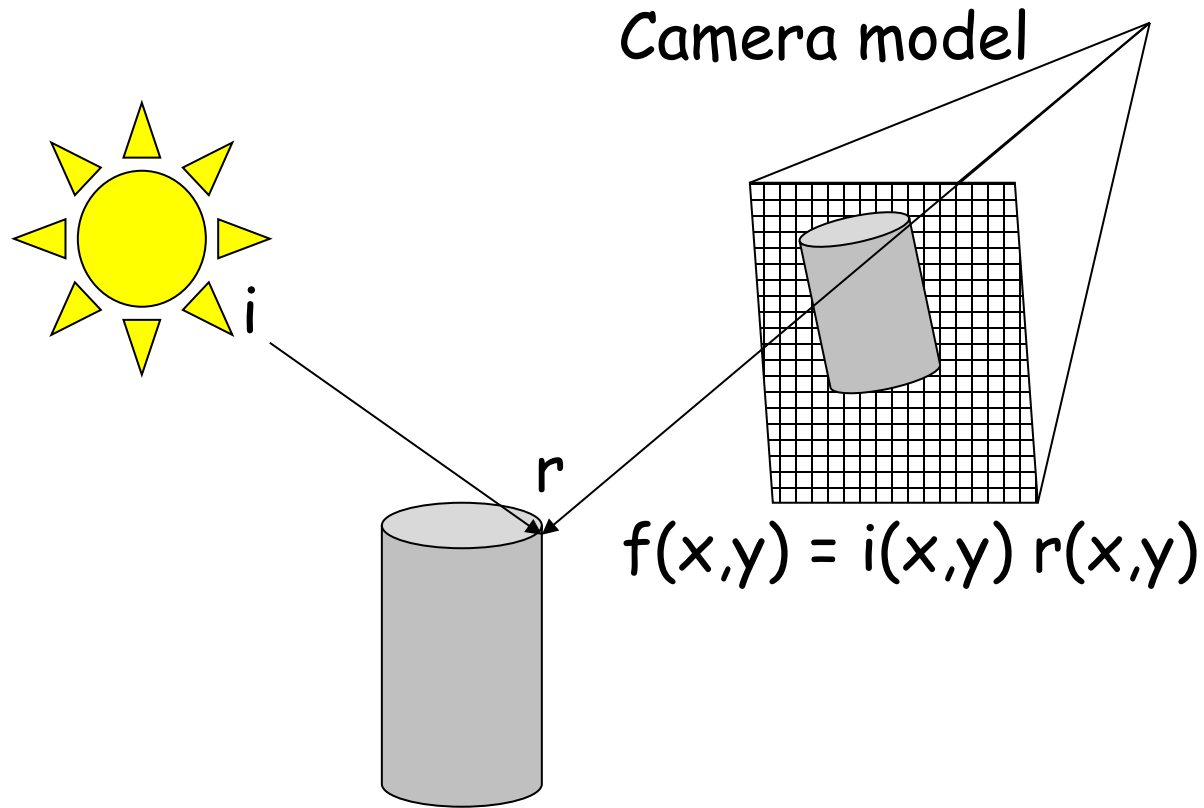
Let's start with a definition. An LRV, or Light Reflectance Value, is a value given to a surface to denote the amount of light it reflects. LRVs are marked on a scale of **1 to 100** depending on the percentage of light reflected. Dark, matt and/or textured surfaces absorb a larger amount of light, and therefore have lower reflectance values.

Light Reflectance Value (LRV) Scale



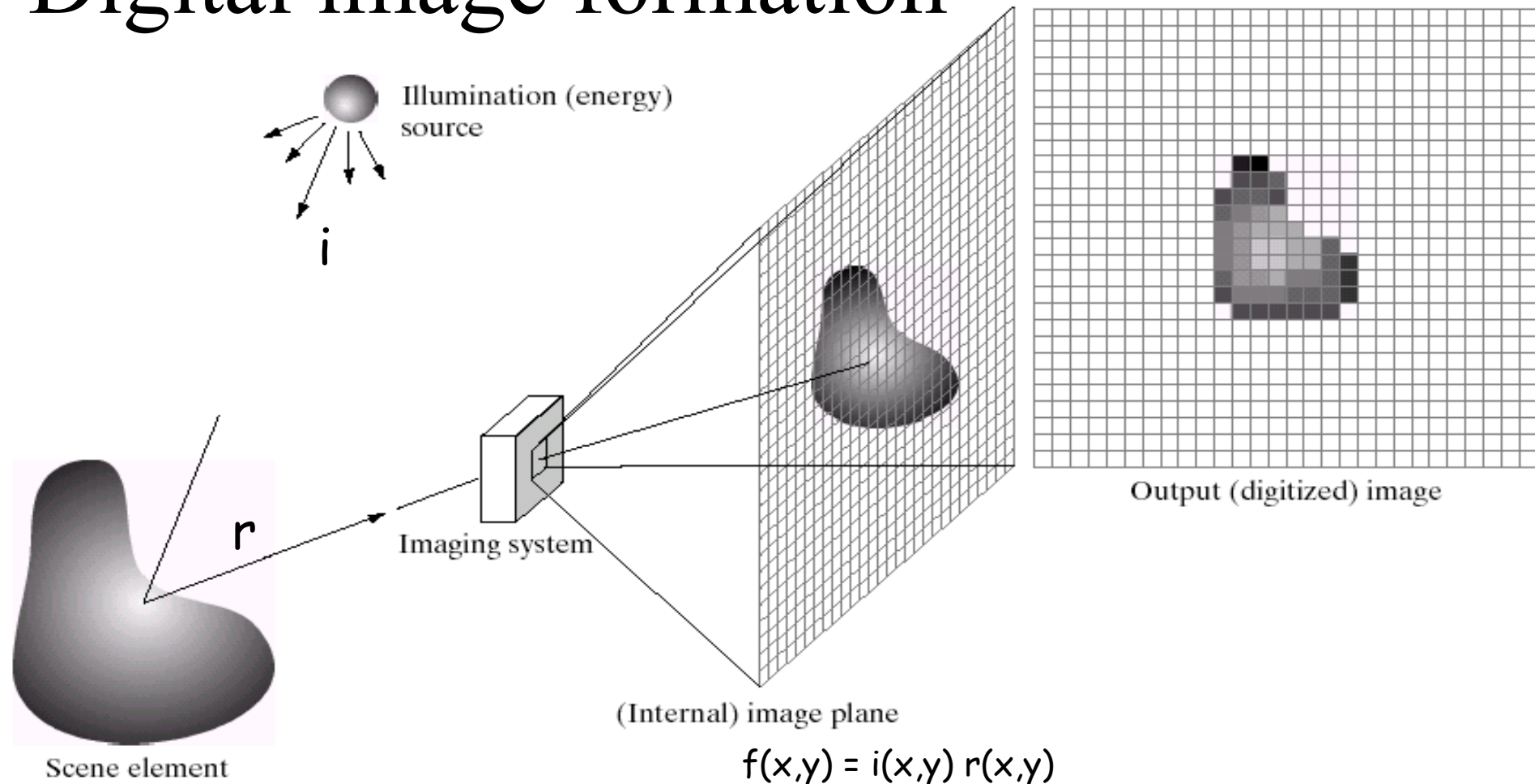
Remember this: The higher the LRV number, the lighter the colour will appear. The lower the LRV number, the darker the colour will appear.

Digital image formation



i and r are continuous functions
thus $f(x,y)$ is continuous

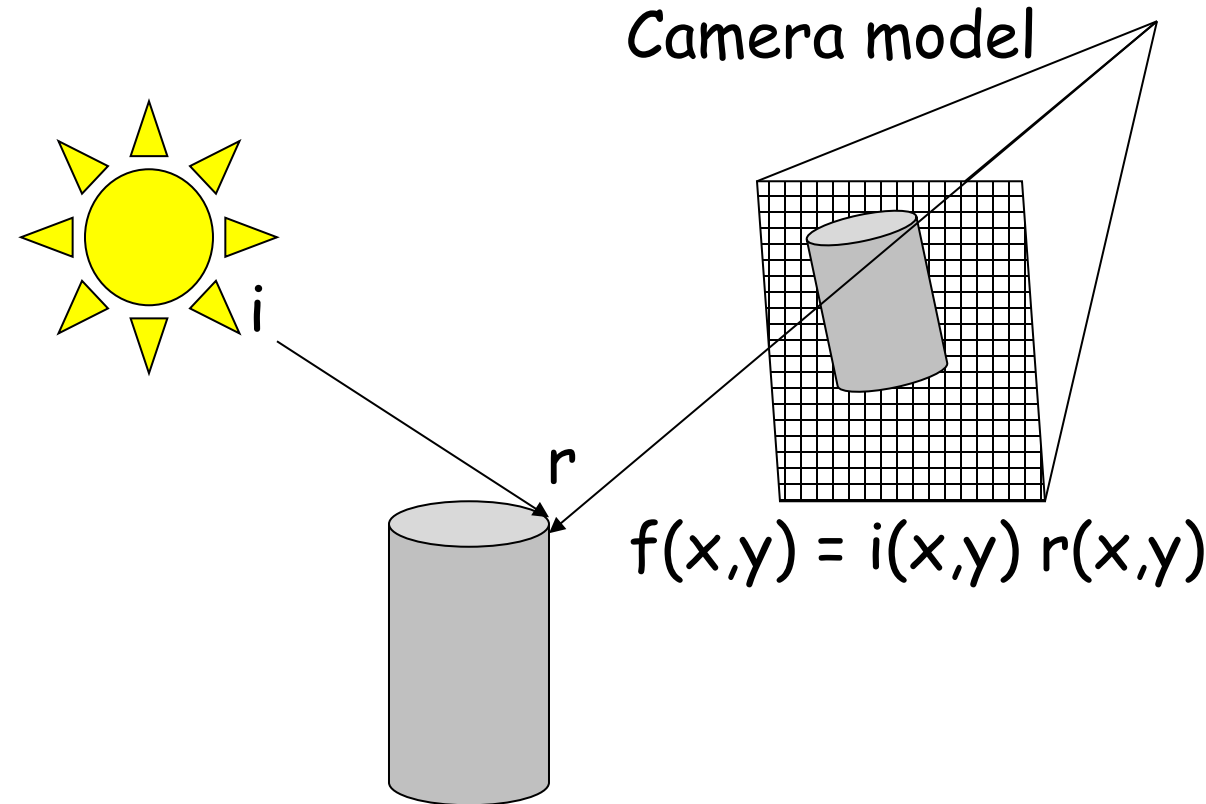
Digital image formation



a b c d e

FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Image resolution



The sampling rate on the image plane is the image resolution.

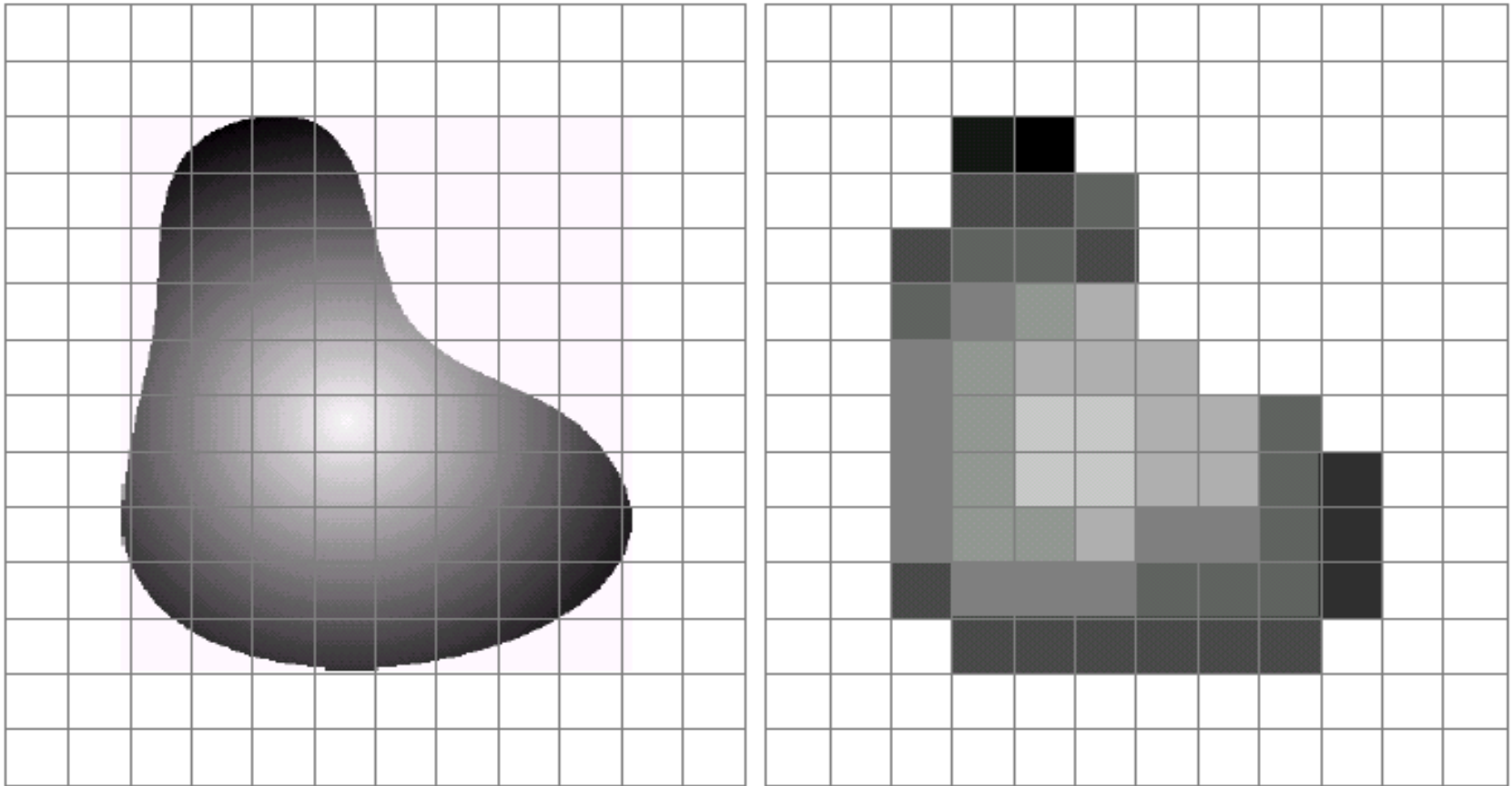
For example, 128 samples along one horizontal line = 128 columns in the 2D image.

100 samples along one vertical line = 100 rows in the 2D image.

The image resolution is 100 x 128 samples or squares (pixels).

The image is represented by a 100 x 128 matrix.

Spatially discretized

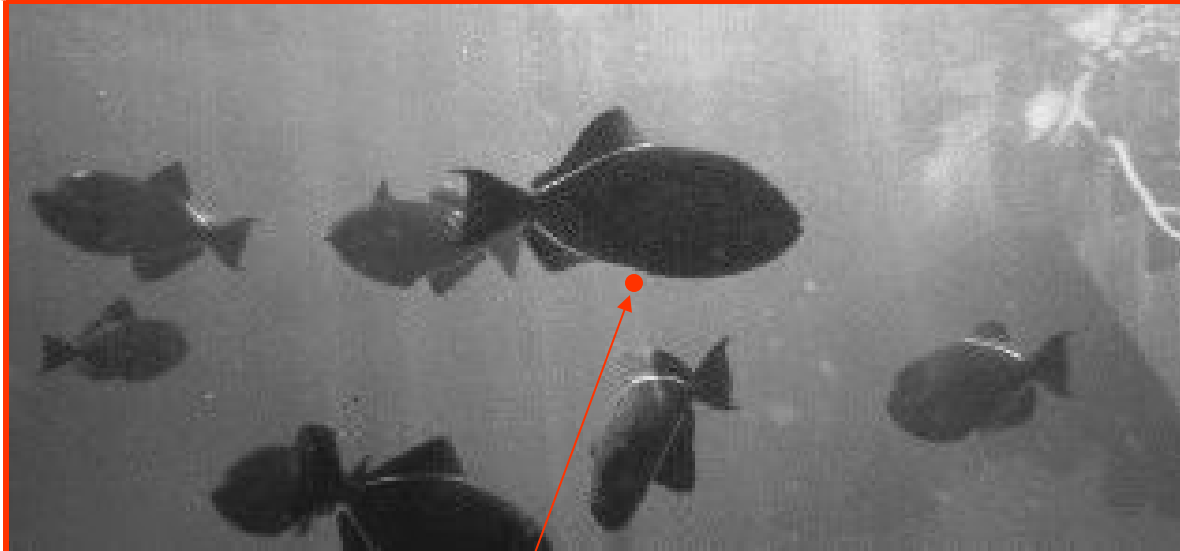


a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

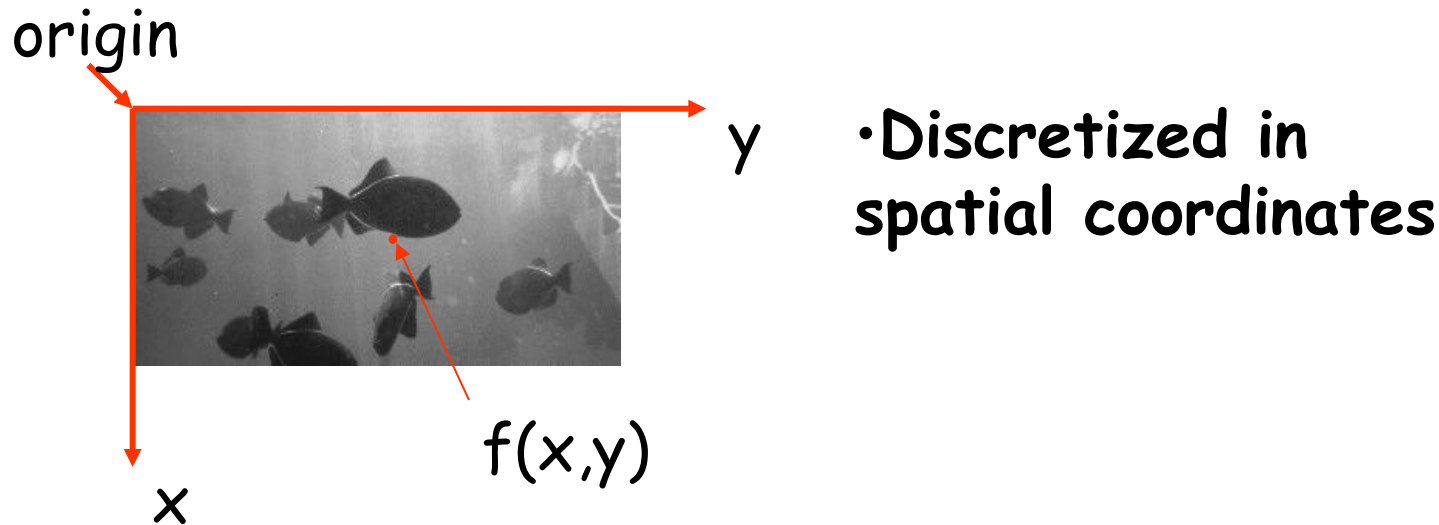
Discretized image representation

origin (0,0)



$f(x,y)$, a 2D function representing image intensity; (x,y) represents a pair of spatial coordinates

Discretized image representation



Like a matrix in Linear Algebra

Individual elements are called:
image elements, picture elements (pixels), or image points.

Discretized image representation

$$f(x,y) \sim \begin{pmatrix} f(0,0) & f(0,1) & . & . & . & f(0,m-1) \\ f(1,0) & f(1,1) & . & . & . & f(1,m-1) \\ . & . & . & . & . & . \\ f(n-1,0) & f(n-1,1) & . & . & . & f(n-1,m-1) \end{pmatrix}$$

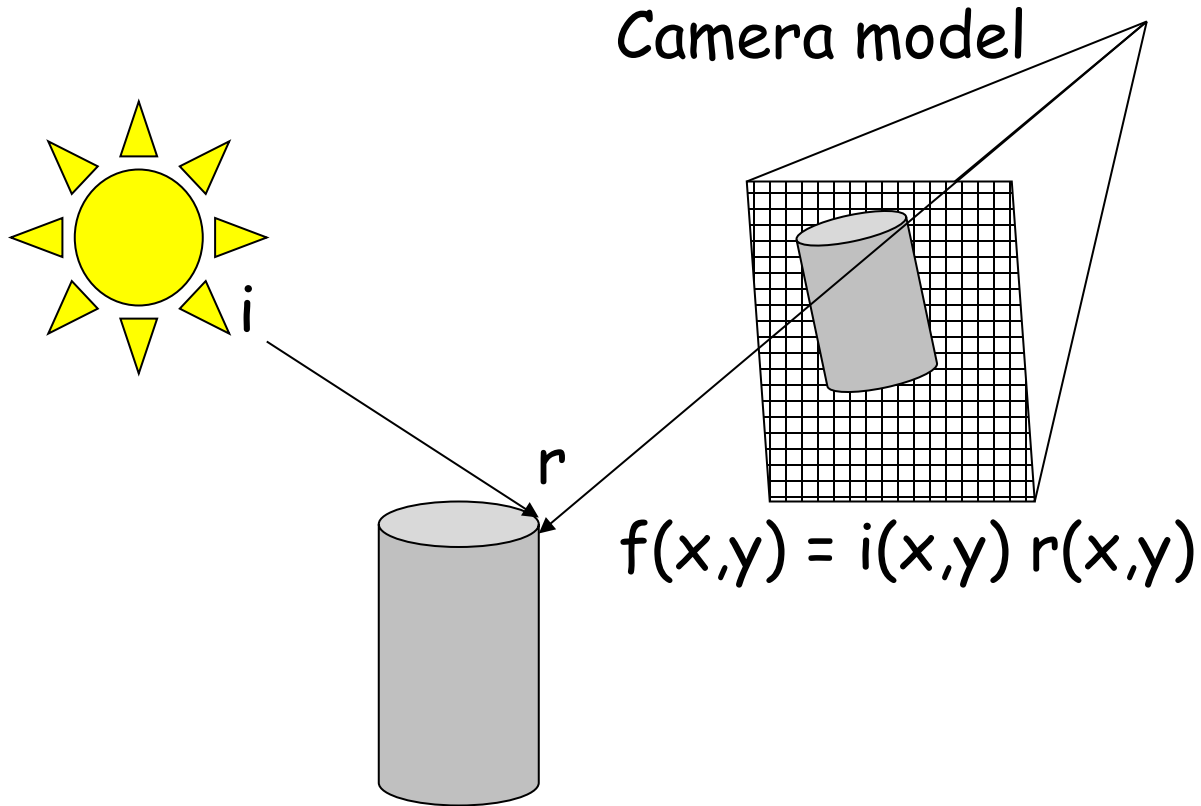
Like a matrix in Linear Algebra with n rows and m columns

Discretized image representation

$f(x, y)$	$f(0, 0)$	$f(0, 1)$	$f(0, 2)$	→ y Dimension 2 (Horizontal), or the 2nd dimension
	$f(1, 0)$	$f(1, 1)$	$f(1, 2)$	
	$f(2, 0)$	$f(2, 1)$	$f(2, 2)$	
↓ x Dimension 1 or the 1st dimension (Vertical)				

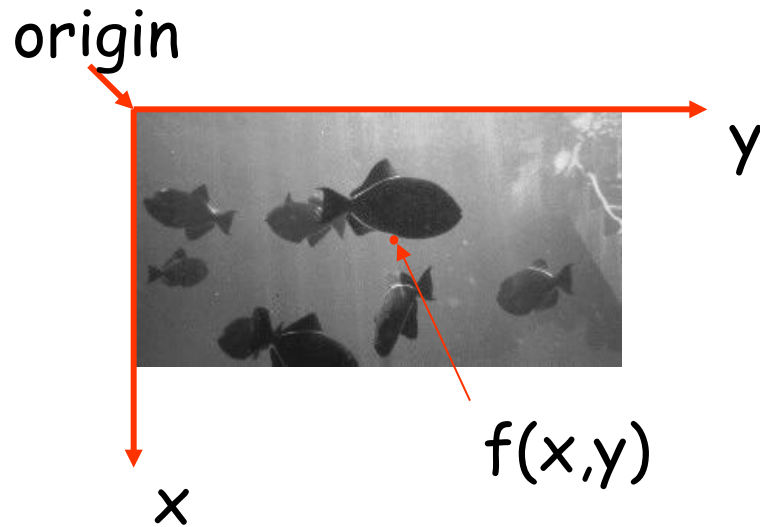
For example, $f(1,0)$ represents one individual element, which is called the image element, picture element (pixels), or image points

Sampling and quantization are affecting the image quality.



i and r are continuous functions
thus $f(x,y)$ is continuous

Discretized image representation



- Discretized in both
 - Spatial coordinates (Sampling)
 - Brightness/Intensity (Quantization)

Discretized image representation

Image can be viewed as a 3D surface with height and different “landscapes”

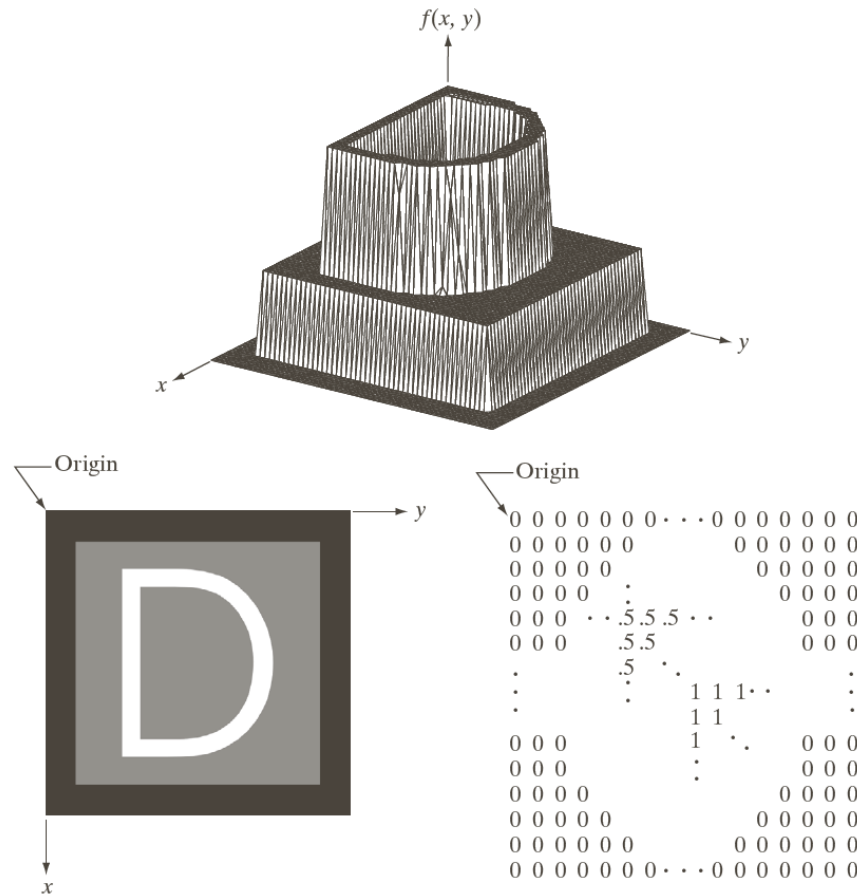


FIGURE 2.18
(a) Image plotted as a surface.
(b) Image displayed as a visual intensity array.
(c) Image shown as a 2-D numerical array (0, .5, and 1 represent black, gray, and white, respectively).

Image Source: Chapter 2, Figure 2.18,

http://www.imageprocessingplace.com/DIP-3E/dip3e_classroom_presentations_downloads.htm

Effects of reducing spatial resolution



a	b	c
d	e	f

More pixelated!

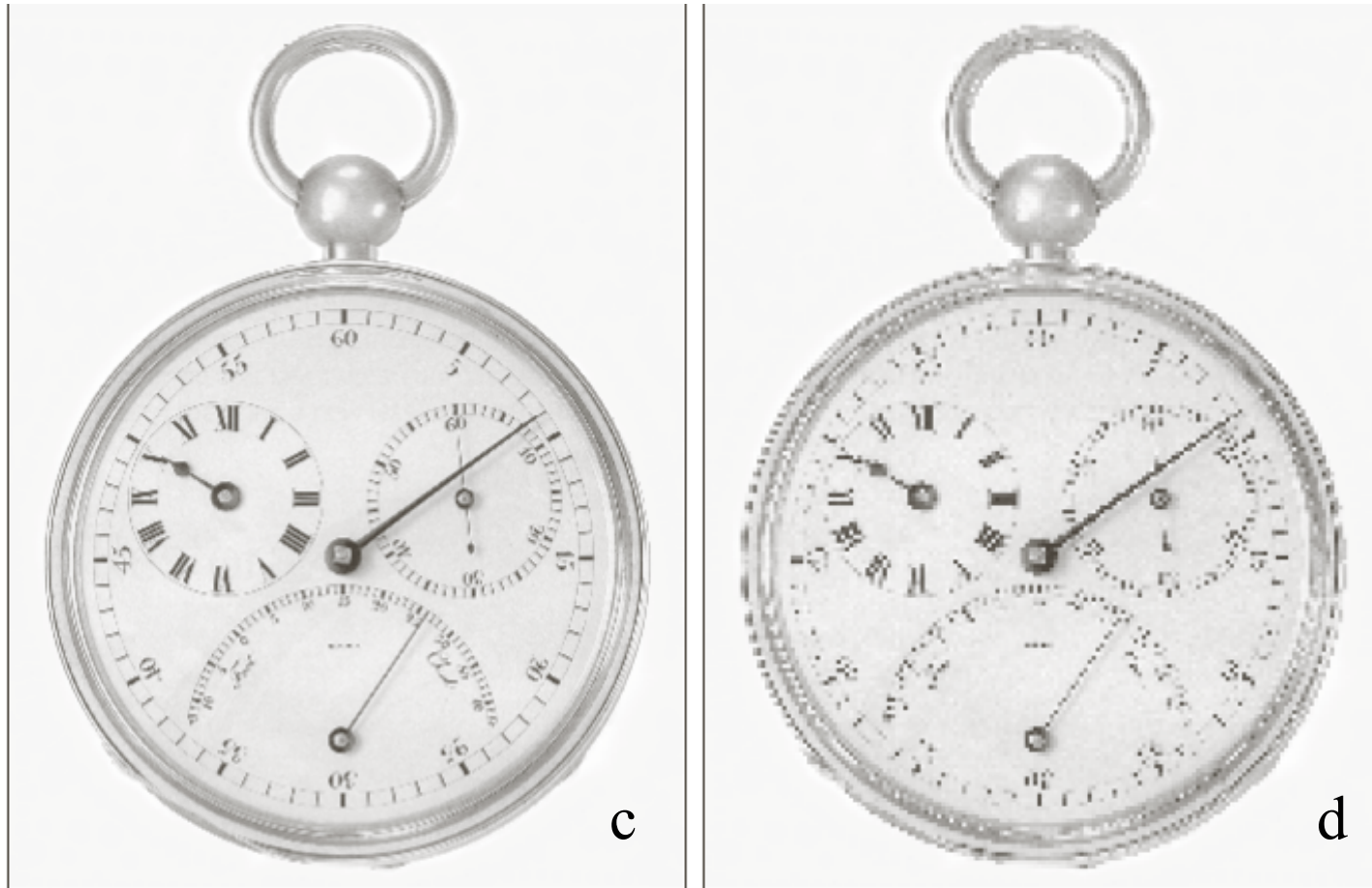
FIGURE 2.20 (a) 1024×1024 , 8-bit image. (b) 512×512 image resampled into 1024×1024 pixels by row and column duplication. (c) through (f) 256×256 , 128×128 , 64×64 , and 32×32 images resampled into 1024×1024 pixels.

Effects of reducing spatial resolution



FIGURE 2.20 Typical effects of reducing spatial resolution. Images shown at: (a) 1250 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi. The thin black borders were added for clarity. They are not part of the data.

Effects of reducing spatial resolution

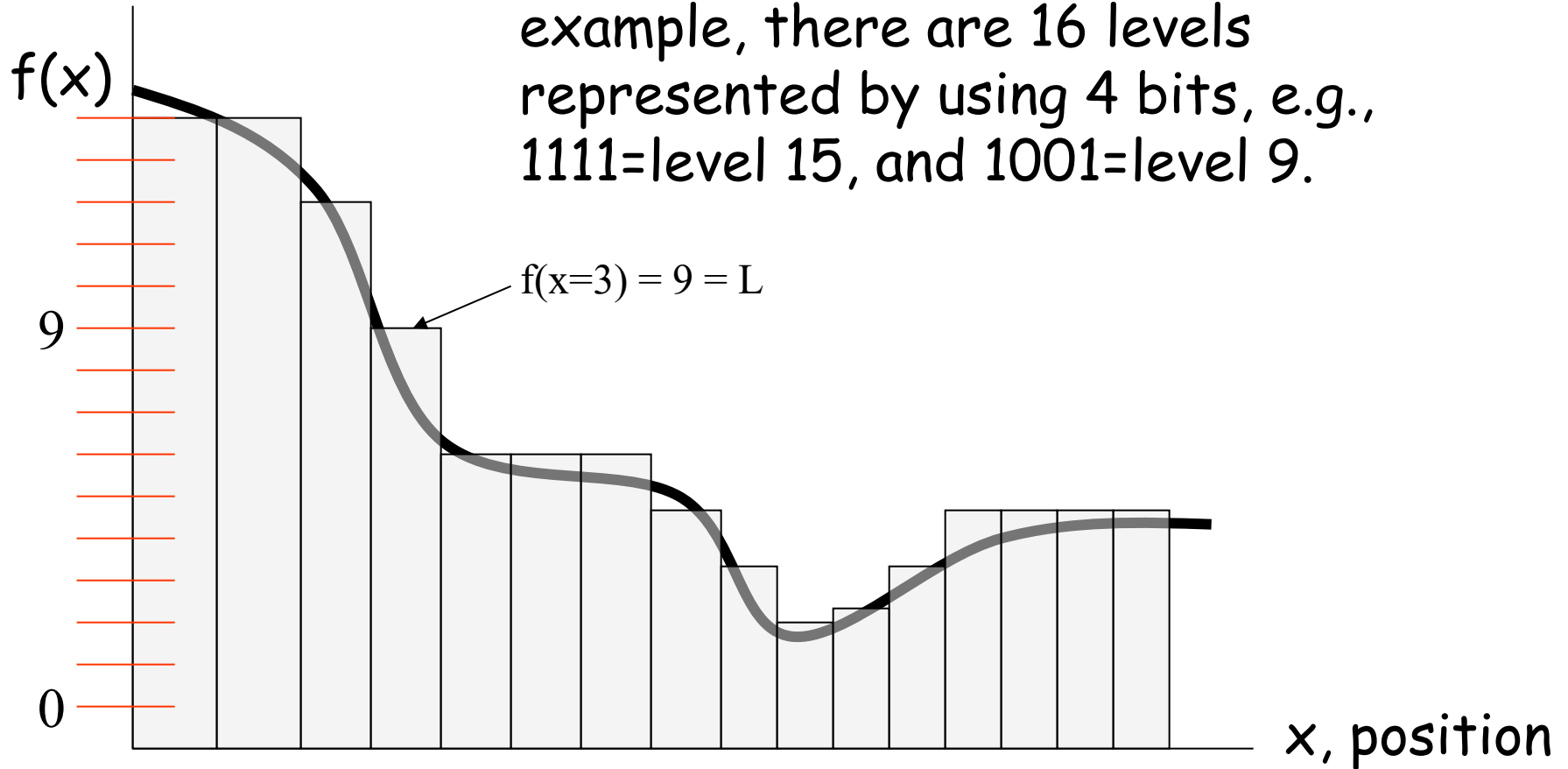


a b
c d

FIGURE 2.20 Typical effects of reducing spatial resolution. Images shown at: (a) 1250 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi. The thin black borders were added for clarity. They are not part of the data.

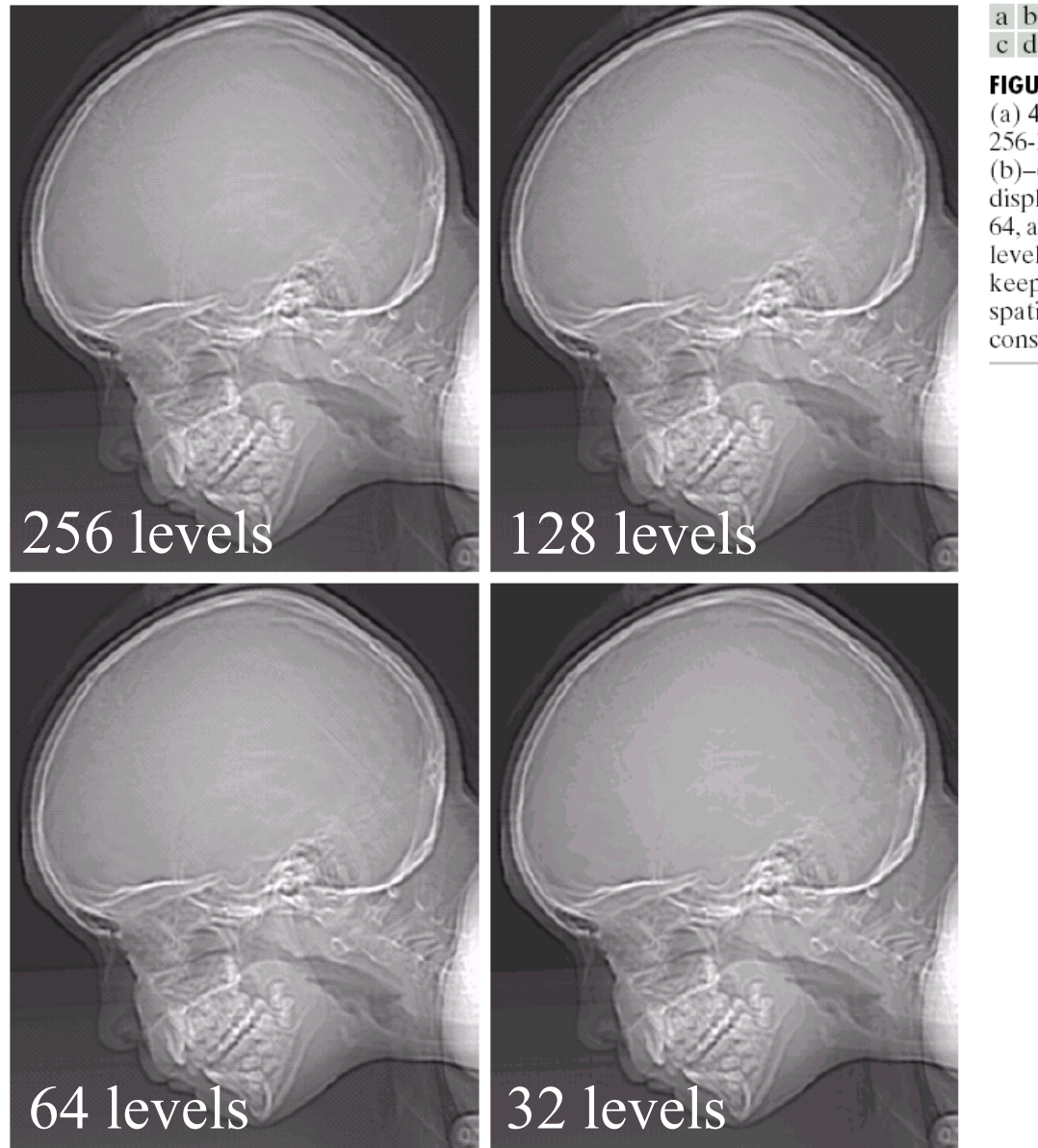
Intensity quantization

We call intensity of a monochrome image, its grey level (L). In this example, there are 16 levels represented by using 4 bits, e.g., 1111=level 15, and 1001=level 9.



Effects of intensity quantization

Image resolution is fixed but the total number of gray levels decreases.



Effects of intensity quantization

e f
g h

FIGURE 2.21

(Continued)

(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)

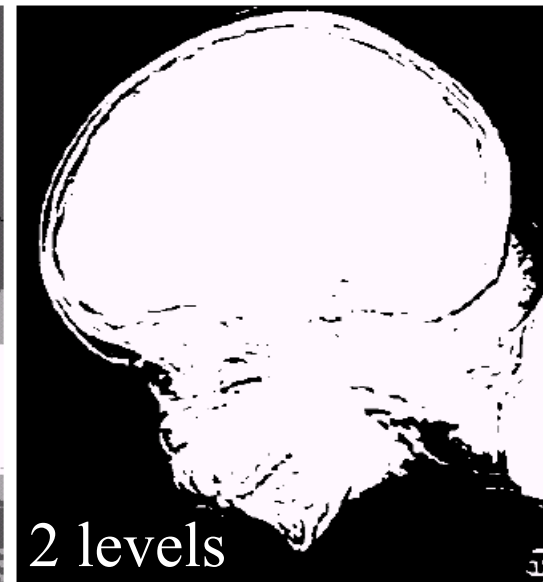
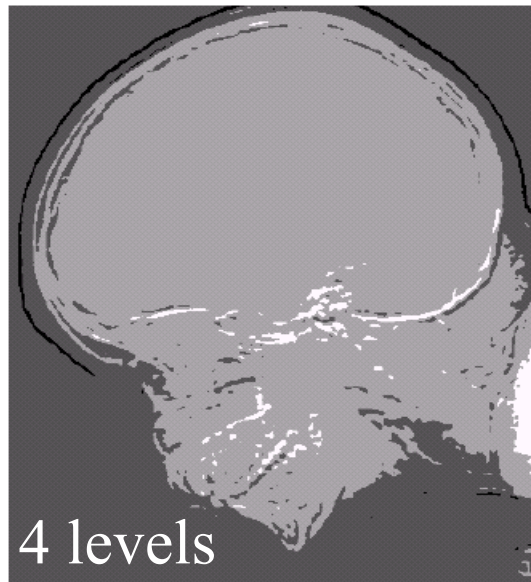
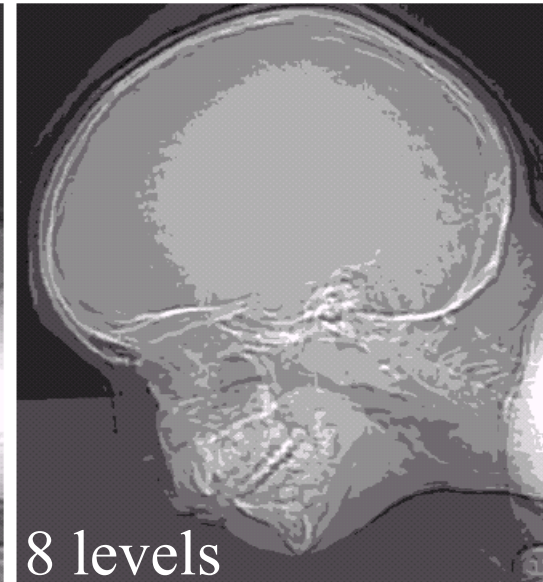
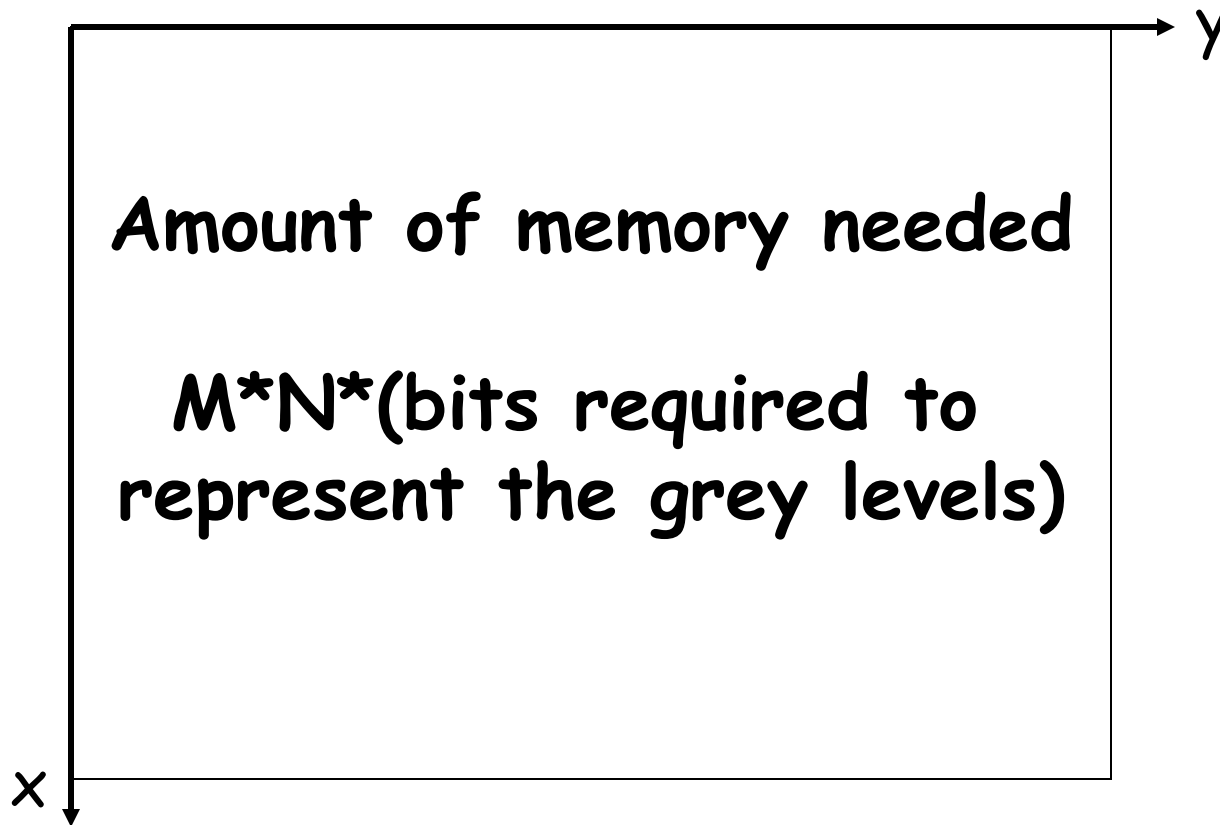


Image resolution is fixed but the number of gray levels decreases.

Storage requirements

M = number of columns,
"Width"



N = number of rows, "height"